Sahaj Bhadja

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Summary

Software Engineer with 2+ years in **distributed systems**, **async pipelines**, and **ETL** on PostgreSQL, Kubernetes, and AWS/GCP; MSCS at USC focused on reliability and low latency.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Database Systems, Operating Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, OOP, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
Built **ETL pipelines** with FastAPI and **PostgreSQL**, processing 5K+ monthly jobs with incremental loads and **CI/CD** at 99%+ reliability. Restructured **async orchestration** with background workers and job-state transitions, processing 1K+ jobs and increasing throughput **3x** under concurrent load.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
Designed **concurrent batch-processing** with multithreading and REST APIs, handling 10K+ daily requests and 1K+ files per batch with fault-tolerant execution. Shipped AWS RDS backend for 150+ users, cutting turnaround **60%** while maintaining stable API performance during peak load through pragmatic tradeoffs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
Engineered a **data-sharing platform** with microservices and **ETL** on AWS Glue and S3, improving response times **30%** and pipeline efficiency **68%**. Built 20+ test suites and CI/CD across Windows, Linux, and macOS, cutting technical debt **20%** while optimizing APIs for high-volume replication workloads.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
Built microservices with WebSockets and REST APIs, **syncing 2M+ records** with real-time updates on relational and document databases. Automated **Docker** and Jenkins **CI/CD** with validation gates, cutting release time **50%** through incremental rollout and automated testing.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
Engineered **data pipelines** with Databricks and Spark on 50K+ events in hourly micro-batches, enabling real-time analytics and sub-200ms API responses. Troubleshot production issues via logging and idempotent webhooks, reducing manual payment handling **70%** while documenting architecture for teams.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++
Data & Backend: PostgreSQL, Redis, ETL Pipelines, FastAPI, Flask, REST APIs, Microservices
Infra & Cloud: Kubernetes, Docker, Terraform, AWS, GCP, CI/CD, Jenkins, Git
Practices: Distributed Systems, Async Systems, Data Integration, React.js, Next.js, Linux, MongoDB, System Design

Projects

Linguistix Code | *Python, AWS, Docker*
Deployed a containerized **data processing API** on AWS ECS/Fargate with CI/CD for high-volume ingestion workloads.

FinNexus Live | *Kubernetes, MongoDB, WebSockets*
Built a platform on **Kubernetes** for 50K+ users and 1M+ transactions with **real-time sync** and low-latency updates.

Big-Brain Live | *GCP, PostgreSQL, React*
Built a cloud **data ingestion pipeline** over 50K+ records on **PostgreSQL** and GCP, cutting retrieval latency **40%**.